

Technical paper

DOGREAT3D: Multimodal 3D reconstruction of complex mechanical parts with diffusion models and geometric refinement strategies

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ABSTRACT

Reconstructing industrial mechanical parts from a single image is challenging when the output must be a CAD-compatible triangular mesh and usable in downstream CAD/CAE workflows after standard cleanup. We propose DOGREAT3D, a CAD-oriented mesh reconstruction pipeline that combines pose-conditioned multi-view diffusion with reliability-aware multimodal fusion and topology-preserving test-time refinement to enforce cross-view geometric consistency. Starting from one image (and an optional short industrial prompt), DOGREAT3D synthesizes posed views, derives geometry cues via cross-view depth alignment, and fuses RGB evidence with depth-derived normals (and optional text priors) through confidence-weighted cross-modal attention. It then initializes a watertight mesh using screened Poisson reconstruction and refines geometry and appearance with a four-stage, topology-preserving schedule that preserves sharp features and sensitive thin structures. On a wheel-hub benchmark of 1500 parts, DOGREAT3D reduces Chamfer Distance by 14.1% and improves F-score@1 mm from 0.453 to 0.526 over the strongest baseline under a unified evaluation protocol. On unseen part categories and on GSO, the geometric components transfer beyond the training distribution, improving F-score@ r from 0.647 to 0.667.

1. Introduction

Fast and reliable 3D reconstruction of industrial mechanical parts is central to many manufacturing-system workflows, including reverse engineering, inspection, remanufacturing, and simulation-driven design iteration [1,2]. In these settings, engineers need geometry that can enter downstream CAD/CAE pipelines as a high-quality mesh representation under tight time and cost constraints, while preserving sharp transitions, openings, and thin structures that strongly affect downstream analyses [3,4]. Yet obtaining dense, clean, and watertight (closed) geometry in real industrial environments remains challenging: occlusions, difficult reflectance, and the practical burden of cleanup and toolchain interoperability continue to be major obstacles [3].

Image-based 3D reconstruction has been widely studied [5,6] and can be roughly grouped into three families. In practice, however, industrial users often have access to only a single view — e.g., from legacy imagery, online catalogs, or on-site capture constrained by time, safety, and accessibility — making multi-view acquisition infeasible. Classical multi-view pipelines based on correspondence and geometric optimization can perform well when photometric cues are stable [7–9], but they often fail on weakly textured or reflective parts, producing missing

structures or over-smoothed surfaces [7]. Neural implicit methods such as DeepSDF [10] and NeuS [11] provide strong function approximators for complex shapes, but typically require substantial training effort and, without additional constraints, do not enforce engineering-critical geometric properties by default. More recently, diffusion-based image-to-3D generators have demonstrated strong shape priors and visually plausible reconstructions, including Zero-1-to-3 [12] and subsequent single-image multi-view diffusion approaches such as Wonder3D [13], supported by the broader diffusion modeling literature [14,15]. Despite this progress, two issues remain particularly acute for industrial parts.

First, visual evidence is often sparse and unreliable: specular highlights, thin structures, self-occlusions, and repeated patterns can destabilize estimation and introduce view-inconsistent details, while standard regularization tends to wash out fine features [7]. Second, engineering use demands more than visual plausibility. A mesh may render well yet remain unusable if it rounds sharp edges, distorts openings or spokes, contains non-watertight artifacts, or exhibits cross-view inconsistencies that undermine geometric coherence and downstream analysis. Throughout this paper, we use *mesh-level CAD-oriented* to

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denote engineering-critical mesh requirements at the mesh stage, including topology correctness (closed/hole-free where needed), preservation of sharp features (e.g., high-dihedral edges), and stable thin structures and openings that are prone to over-smoothing. These requirements motivate a hybrid design that couples generative priors with geometry-driven refinement in a controlled and reproducible manner, improving not only appearance but also engineering-relevant geometric fidelity. Concretely, we aim to (i) extract geometry cues that remain consistent across synthesized views (mitigating monocular scale ambiguity and view-dependent artifacts), (ii) integrate multimodal evidence in a reliability-aware way (down-weighting cues corrupted by specularities or weak texture), and (iii) refine geometry with topology-preserving objectives that retain sharp features and sensitive structures rather than smoothing them away. We propose DOGREAT3D, a mesh reconstruction pipeline tailored to industrial mechanical parts. Given a single input image (and an optional short industrial prompt), DOGREAT3D first synthesizes a fixed set of posed views using a Zero123/Zero123Plus-style multi-view diffusion backbone [12,16,17] with ControlNet conditioning [18]. In many industrial settings, camera intrinsics are available, and object pose can be approximately controlled (e.g., via a fixture or turntable) or coarsely initialized; we leverage this to place synthesized views on a canonical orbit with consistent extrinsics. When these conditions are imperfect, performance degrades gracefully and we discuss practical mitigation strategies in Section 5. We then align monocular depth across views to obtain geometry-consistent cues and derive per-view normals. A confidence-aware fusion module integrates RGB evidence, depth-derived normals, and optional text guidance via confidence-weighted cross-modal attention, building on CLIP-style vision–language representations when text is provided [19]. Finally, DOGREAT3D initializes an explicit mesh via screened Poisson reconstruction [20,21] and applies a topology-preserving, four-stage test-time refinement schedule that progressively enforces flow-guided photometric alignment, depth reprojection consistency, and normal alignment with feature-preserving regularization, followed by multi-view color fitting.

The main contributions are:

- A mesh-level engineering-oriented reconstruction pipeline that combines diffusion-based posed view synthesis with topology-preserving staged refinement, improving cross-view geometric consistency and sharp-feature fidelity with stable, reproducible refinement behavior.
- A confidence-aware multimodal fusion mechanism for industrial imagery. By weighting RGB and normal evidence (and an optional text prior) according to reliability, the method is more robust to specularities, weak texture, and thin structures.
- Manufacturing-oriented evaluation and downstream feasibility. We establish a wheel-hub benchmark with ablations and robustness analyses, evaluate cross-domain performance on GSO, and demonstrate file-level interoperability (STEP export) and a solvable linear static analysis workflow in ANSYS after standard mesh cleanup (e.g., removing isolated components, resolving non-manifold artifacts, and minor hole filling) as a feasibility study rather than a certification-grade analysis.

The remainder of this paper is organized as follows. Section 2 reviews relevant literature. Section 3 describes the proposed DOGREAT3D method. Section 4 presents experimental validations and comparisons. Section 5 discusses findings, implications, and limitations. Finally, Section 6 summarizes conclusions and outlines future research directions.

2. Related works

2.1. Image-based 3D reconstruction

Classical image-based reconstruction pipelines recover geometry from multi-view imagery via camera pose estimation and triangulation.

Structure-from-Motion (SfM) estimates camera motion and sparse structure from unordered images [8,22], and Multi-View Stereo (MVS) further densifies geometry by multi-view matching [7,23]. While effective in many scenarios, these approaches can degrade on industrial parts due to weak textures, repeated patterns, and specular highlights that destabilize correspondence [9,24]. Beyond purely geometric pipelines, learning-based reconstruction methods aim to infer 3D shape from limited views, often by leveraging implicit representations and differentiable rendering. Neural Radiance Fields (NeRF) [25] demonstrated that neural volumetric rendering can produce high-fidelity view synthesis and geometry cues, inspiring a broad family of image-conditioned 3D reconstruction methods. In manufacturing-oriented reverse engineering, reconstruction is commonly followed by steps that improve engineering usability, such as segmentation, surface/primitive fitting, topology repair, and mesh cleanup for downstream CAD/CAE. This perspective emphasizes that practical deployment depends not only on geometric accuracy but also on representation quality (e.g., watertightness and clean topology) and on preserving features relevant to inspection and simulation.

2.2. Deep generative 3D modeling and diffusion-based view synthesis

Early data-driven paradigms explored learning-based shape priors for 3D generation and reconstruction, including generative modeling in the presence of sparse observations and complex geometry [6,26]. With the rapid development of diffusion models [14,15], diffusion-based image-to-3D pipelines have shown powerful priors that can produce visually plausible reconstructions from limited inputs. In particular, diffusion can be used to synthesize posed multi-view observations from a single image, which can then be converted into explicit 3D geometry. Representative methods include Zero-1-to-3 [12] and subsequent single-image multi-view diffusion approaches such as Wonder3D [13]. In parallel, general-purpose 3D generative systems such as Point-E [27] and Shap-E [28] demonstrate the potential of text-conditioned 3D generation, further motivating the use of learned priors for downstream reconstruction under limited evidence. However, producing engineering-friendly geometry typically requires additional reconstruction and refinement to control topology and preserve sharp features beyond visual plausibility. Overall, these advances suggest a practical direction: generating consistent multi-view cues with strong priors and then enforcing geometric consistency through explicit reconstruction and refinement.

2.3. Hybrid pipelines: Learned priors with geometric optimization

A practical trend is to combine learned priors with explicit geometric constraints and optimization to improve reconstruction quality under imperfect observations. On the one hand, deep models can enhance feature extraction and facilitate multi-source fusion, e.g., integrating heterogeneous sensing modalities for improved 3D modeling [29,30]. On the other hand, geometric consistency constraints remain essential for stabilizing reconstruction, especially when correspondence is unreliable or view synthesis introduces inconsistencies. Classical image-based reconstruction studies highlight that matching and triangulation can be brittle under weak texture, repeated patterns, and challenging reflectance [7–9], motivating the use of additional regularization and refinement beyond purely learning-based outputs. In post-processing, learning-based completion or inpainting has also been used to repair holes and improve completeness of 3D models [26,31]. Our work follows this hybrid philosophy but focuses on industrial parts where engineering-critical mesh structure (sharp features, openings, and stable thin elements) and topology-preserving refinement are crucial.

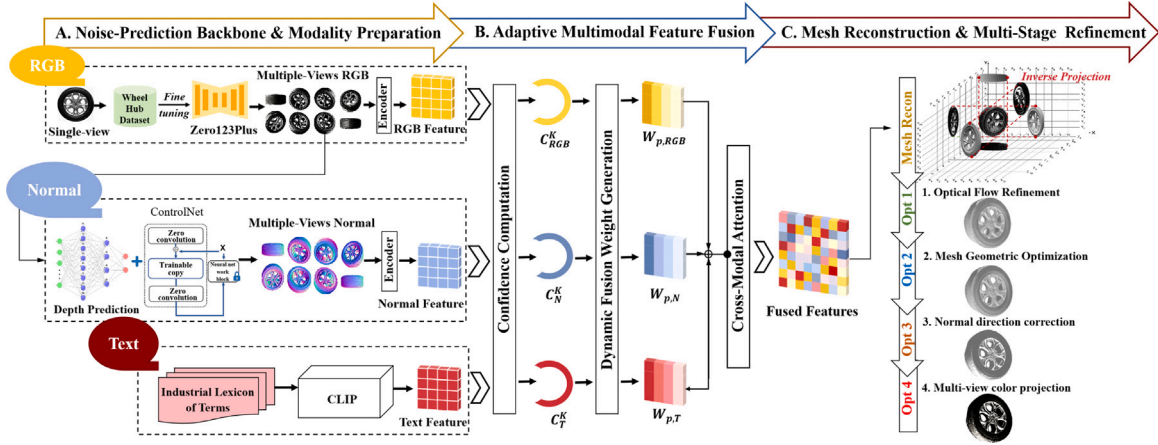


Fig. 1. Overview of DOGREAT3D. (A) **Noise-prediction backbone and modality preparation:** from a single input image (and an optional prompt), we synthesize posed multi-view RGB images, align monocular depth across views, and derive depth-based normal cues. (B) **Adaptive multimodal feature fusion:** we estimate modality confidences and fuse RGB, normals, and optional text priors via confidence-weighted cross-modal attention to obtain fused features and refined normals. (C) **Mesh reconstruction and multi-stage refinement:** we initialize a watertight mesh via screened Poisson reconstruction and refine it using a four-stage, topology-preserving schedule (Opt 1–4): (1) flow-guided photometric alignment, (2) depth reprojection consistency, (3) normal alignment with feature-preserving regularization, and (4) multi-view color projection/fitting. (Opt = optimization; Recon = reconstruction; $c_p^{(-)}$ denotes confidence scores; $w_{p,(-)}$ denotes fusion weights.)

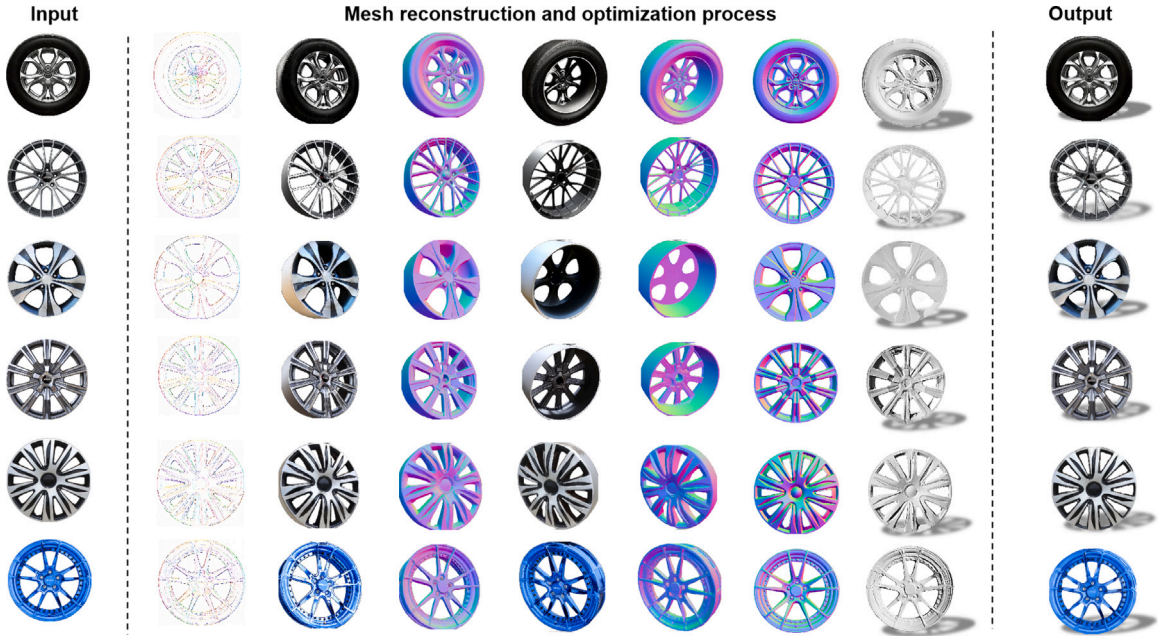


Fig. 2. Multi-stage mesh refinement workflow showing enhanced geometric detail preservation for mechanical parts.

3. Methodology

3.1. Overview, problem setup, and notation

DOGREAT3D is a three-stage pipeline that converts a single input view into an explicit CAD-oriented mesh representation, with stable and reproducible test-time refinement (Figs. 1–2). Given an input image (and an optional short industrial prompt), DOGREAT3D: (i) synthesizes a fixed set of posed multi-view RGB observations with a pose-conditioned diffusion backbone, predicts per-view monocular depth, and aligns depth across views to derive geometry-consistent cues, including depth-based normals; (ii) performs confidence-aware multimodal fusion that adaptively weights RGB evidence, depth-derived normals, and optional text priors to predict refined normals and fused features; and (iii) initializes an explicit mesh via screened Poisson

reconstruction and applies a four-stage, topology-preserving test-time refinement schedule that progressively enforces flow-guided photometric alignment, depth reprojection consistency, and normal alignment with feature-preserving regularization, followed by multi-view color fitting. Importantly, test-time refinement optimizes only mesh variables (geometry and appearance) and does not backpropagate through the diffusion generator, decoupling generation from optimization and improving deployment robustness.

DOGREAT3D targets engineering-oriented mesh reconstruction of industrial mechanical parts under limited visual evidence. Given a single RGB image $I^0 \in \mathbb{R}^{H \times W \times 3}$ and an optional industrial prompt s , the method outputs a textured mesh $\hat{M} = (\hat{V}, \hat{F}, \hat{T})$, where $\hat{V}$ and $\hat{F}$ denote vertices and faces, and $\hat{T}$ denotes surface appearance. In our implementation, appearance is represented by per-vertex colors; the same pipeline can be used with UV textures.

Table 1

Module interface contract (stage inputs/outputs). $I^1 = I^0$. $(\cdot)_c$: camera-frame normals.

Module	Input $\rightarrow$ Output
Diffusion + raw modalities	$(I^0, s, \mathbf{K}, \{\mathbf{E}^k\})$ $\rightarrow \{I^k, D_{\text{raw}}^k\}_{k=1}^{N_v}$
Depth alignment + cues	$\{D_{\text{raw}}^k, I^k, \mathbf{K}, \mathbf{E}^k, f_{i \rightarrow j}\}$ $\rightarrow \{\tilde{D}^k, c_{p,D}^k, \tilde{N}_c^k\}$
Confidence + fusion	$\{I^k, \tilde{D}^k, \tilde{N}_c^k, f_{i \rightarrow j}, s\}$ $\rightarrow \{F_{\text{fused}}^k, \tilde{N}_c^k, c_{p,RGB}^k, c_{p,N}^k, c_S\}$
Mesh init + refinement	$\{\tilde{D}^k, \tilde{N}_c^k, I^k, c_{p,RGB}^k, c_{p,N}^k, c_{p,D}^k\}$ $\rightarrow M_{\text{init}} \rightarrow \hat{M}$

In manufacturing workflows, visually plausible meshes are insufficient if sharp transitions, openings, or thin structures are rounded, displaced, or collapsed. We therefore treat mesh-level CAD-oriented reconstruction as preserving sharp features and locally low-curvature primitives *when supported by the available evidence*. During refinement, the mesh topology $\mathcal{F}$ is fixed and only vertex positions and appearance are optimized, preventing uncontrolled topological drift at test time. This fixed-topology design stabilizes refinement under imperfect multi-view cues and yields reproducible optimization behavior. Operationally, we enforce engineering-critical structure by an edge-aware dihedral-angle regularizer: edges that are sharp in M_{init} are constrained to remain sharp, while initially smooth edges are softly regularized to preserve low-curvature primitives (Section 3.4.6).

3.1.1. Deployment setting and assumptions

DOGREAT3D assumes a calibrated or semi-calibrated acquisition setting where camera intrinsics are known and the input image can be associated with a canonical orbit view. This assumption matches common industrial capture workflows (fixture/turntable stations) and enables reproducible geometry reasoning across synthesized views. When calibration or pose references are imperfect, we use coarse initialization (e.g., discrete yaw hypotheses) and report sensitivity to such mismatches in Section 4. Concretely, we assume:

- **Intrinsics:** a single intrinsic matrix $\mathbf{K}$ is available (or can be calibrated once for a capture station).
- **Pose reference:** the input view is mapped to the nearest canonical orbit view. This mapping is reliable when capture is controlled (fixed camera with repeatable placement), and can be approximated by a coarse pose initialization (e.g., discrete yaw hypothesis selection) when placement is less controlled.
- **Robustness:** downstream modules operate given $\mathbf{K}$ and $\{\mathbf{E}^k\}$; we analyze sensitivity to pose/orbit mismatches in Section 4.

We generate N_v posed views on a canonical orbit. To keep notation consistent with the single-input setting, we treat the input as the reference orbit view and set $I^1 \triangleq I^0$ with a corresponding reference pose $\mathbf{E}^1$. Views $k \geq 2$ are synthesized conditioned on the remaining orbit poses $\{\mathbf{E}^k\}_{k=2}^{N_v}$. Cross-view correspondences are denoted by $f_{i \rightarrow j}$, which maps pixels from view i to view j (e.g., optical flow estimated between posed RGB views) and is used for cross-view warping and consistency.

3.1.2. Module interface contract

For reproducibility and implementation clarity, we summarize the stage-wise interfaces (inputs/outputs and coordinate frames) in Table 1. This “contract” makes explicit what each module consumes and produces, and how the outputs are used downstream.

3.1.3. Camera convention and validity masks

The posed views are defined by a fixed set of target extrinsics $\{\mathbf{E}^k\}_{k=1}^{N_v}$ on a canonical orbit around the object. In training and evaluation under the rendered-data protocol, these extrinsics are known by construction.

The world-to-camera mapping is

$$\mathbf{x}_c^k = \mathbf{R}^k \mathbf{x}_w + \mathbf{t}^k, \quad \mathbf{p} = \Pi(\mathbf{K} \mathbf{x}_c^k), \quad \Pi([x, y, z]^\top) = [x/z, y/z]^\top. \quad (1)$$

We represent image locations by continuous coordinates $\mathbf{p} = (p_x, p_y) \in \mathbb{R}^2$ produced by projection and flow warping. Image, depth, normal, and confidence values are queried by bilinear sampling at $\mathbf{p}$ and gated by an in-bounds indicator

$$\mathbb{I}_{\text{in}}(\mathbf{p}) = \mathbb{I}(0 \leq p_x \leq W-1 \wedge 0 \leq p_y \leq H-1). \quad (2)$$

For differentiable rendering, $\mathbb{I}_{\text{vis}}^k(\mathbf{p}) \in \{0, 1\}$ denotes the z-buffer visibility mask in view k . All pixel-based losses and confidence pooling use

$$\mathbb{I}_{\text{valid}}^k(\mathbf{p}) = \mathbb{I}_{\text{in}}(\mathbf{p}) \cdot \mathbb{I}(\tilde{D}^k(\mathbf{p}) > \epsilon_D) \cdot \mathbb{I}(\tilde{D}^k(\mathbf{p}) \text{ is finite}). \quad (3)$$

3.1.4. Indexing and modality notation

The view index is $k \in \{1, \dots, N_v\}$, pixel coordinate $\mathbf{p}$, and token index t on a downsampled grid used by the fusion module. Fusion modalities are indexed by $m \in \{\text{RGB}, N, S\}$. Aligned depth is $\tilde{D}^k$ and raw depth is D_{raw}^k . Depth is not treated as a fusion modality; its reliability $c_{p,D}^k$ is used only in geometric stages. Pixel-level confidences $c_{p,\cdot}^k \in [0, 1]$ are defined for $\{\text{RGB}, N, D\}$. Text reliability is global and injected as $\tilde{c}_{t,S}^k \equiv c_S$ when enabled. Fusion weights $\omega_{t,m}^k$ satisfy $\sum_{m \in \mathcal{M}(s)} \omega_{t,m}^k = 1$, where $\mathcal{M}(s) = \{\text{RGB}, N\}$ if s is absent and $\mathcal{M}(s) = \{\text{RGB}, N, S\}$ otherwise. When supervising pixel-space losses, token-space fusion quantities are aligned to pixel coordinates by bilinear sampling/upsampling consistent with the warping operator.

We instantiate the multi-view generator using Zero123Plus+ ControlNet. Fusion and test-time refinement are downstream modules that operate given camera intrinsics, an orbit pose reference, and cross-view correspondences.

3.2. Noise-prediction diffusion backbone and modality preparation

Camera intrinsics/extrinsics follow the fixed camera protocol described in Section 3.1.1 and Table A.10, and are used to condition the diffusion model on target viewpoints. We sample N_v views on a fixed-radius orbit with uniformly spaced azimuths and fixed elevation. A single intrinsic matrix $\mathbf{K}$ is used for all views. The generator takes $\{\mathbf{E}^k\}$ and $\mathbf{K}$ as inputs and outputs posed images $\{I^k\}$.

The diffusion backbone is fine-tuned on the wheel-hub dataset with the DDPM noise-prediction objective:

$$\mathcal{L}_{\text{noise}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left\| \epsilon - \epsilon_\theta(\mathbf{x}_t, t) \right\|_2^2. \quad (4)$$

A pretrained estimator f_{depth} produces raw monocular depth D_{raw}^k for each view [32]. Because monocular depth is up-to-scale and can vary across views, we estimate per-view affine parameters (a_k, b_k) and define $\tilde{D}^k(\mathbf{p}) = a_k D_{\text{raw}}^k(\mathbf{p}) + b_k$. Alignment minimizes 3D disagreement in the world frame under flow correspondences and the camera protocol.

Let $f_{i \rightarrow j}$ be optical flow between views (i, j) . For a pixel $\mathbf{p}$ in view i , its correspondence in view j is $\mathbf{p}_j = \mathbf{p} + f_{i \rightarrow j}(\mathbf{p})$. Back-projection to the camera frame uses

$$\mathbf{x}_c^k(\mathbf{p}) = \tilde{D}^k(\mathbf{p}) \mathbf{K}^{-1} [p_x, p_y, 1]^\top, \quad (5)$$

and conversion to the world frame uses

$$\mathbf{x}_w^k(\mathbf{p}) = (\mathbf{R}^k)^\top (\mathbf{x}_c^k(\mathbf{p}) - \mathbf{t}^k). \quad (6)$$

Positivity is enforced by $a_k = \exp(\tilde{a}_k)$. With gauge fixing $\tilde{a}_1=0$ and $b_1=0$, we solve $\{\tilde{a}_k, b_k\}$ by

$$\min_{\{\tilde{a}_k, b_k\}} \sum_{(i,j) \in \mathcal{P}} \sum_{\mathbf{p} \in \mathcal{I}_{i \rightarrow j}} \rho(\|\mathbf{x}_w^i(\mathbf{p}) - \mathbf{x}_w^j(\mathbf{p}_j)\|_2) + \lambda_a \sum_{k>1} \tilde{a}_k^2 + \lambda_b \sum_{k>1} b_k^2, \quad (7)$$

where $\mathcal{P}$ is an adjacent view-pair set and $\rho(\cdot)$ is defined in Eq. (11). The valid set $\Omega_{i \rightarrow j}$ keeps pixels with in-bounds sampling in both views, finite aligned depths, forward-backward flow consistency, a loose photometric gate, and non-saturation; thresholds are provided in Table A.10.

Saturated pixels are identified by

$$\text{sat}^k(\mathbf{p}) = \mathbb{I}(\max(I^k(\mathbf{p})) > \xi). \quad (8)$$

We initialize $\tilde{a}_k$ and b_k to zero, clamp aligned depth by $\tilde{D}^k \leftarrow \max(\tilde{D}^k, \epsilon_D)$, and solve Eq. (7) using L-BFGS with uniform subsampling of $\Omega_{i \rightarrow j}$.

We compute depth-derived normals in the camera frame by finite differences on back-projected 3D points. Back-projection uses

$$\mathbf{x}_c^k(u, v) = \tilde{D}^k(u, v) \mathbf{K}^{-1}[u, v, 1]^T, \quad (9)$$

where (u, v) indexes integer pixel centers. We compute initial normals by local cross products:

$$\tilde{\mathbf{n}}_c^k(u, v) = \text{normalize}((\mathbf{x}_c^k(u + \Delta_u, v) - \mathbf{x}_c^k(u, v)) \times (\mathbf{x}_c^k(u, v + \Delta_v) - \mathbf{x}_c^k(u, v))). \quad (10)$$

where $(\Delta_u, \Delta_v) = (1, 1)$ for interior pixels and backward differences are used at image boundaries. Stacking $\tilde{\mathbf{n}}_c^k(u, v)$ yields $\tilde{N}_c^k$.

If s is provided, a CLIP text embedding $V_S \in \mathbb{R}^{512}$ is extracted as a global semantic prior [19]. This prior is used only as an auxiliary modality in fusion (Section 3.3.2) and is gated by a text reliability score (Section 3.3.1) to prevent irrelevant prompts from dominating geometric cues. If s is absent, the text branch is disabled by modality masking in Section 3.3.2.

3.3. Adaptive multimodal feature fusion

3.3.1. Confidence modeling with one-step normal-consistency update

Industrial parts often exhibit specularities, weak texture, and thin structures, which can break cross-view photometric consistency and destabilize depth-derived geometry. We therefore estimate pixel-level confidences for RGB and normals, and compute a depth reliability signal used only in geometric stages.

We adopt a Charbonnier penalty to reduce the influence of outliers:

$$\rho(x) = \sqrt{x^2 + \epsilon_p^2}. \quad (11)$$

For RGB residuals $\Delta \mathbf{I} \in \mathbb{R}^3$, we use $\rho(\|\Delta \mathbf{I}\|_1)$. Photometric consistency is evaluated under flow correspondences:

$$e_{\text{RGB}}^k(\mathbf{p}) = \text{median}_{j \in \mathcal{N}(k)} \rho(\|I^k(\mathbf{p}) - I^j(\mathbf{p} + f_{k \rightarrow j}(\mathbf{p}))\|_1). \quad (12)$$

Saturation is handled by $\text{sat}^k(\mathbf{p})$ in Eq. (8). The RGB confidence is

$$c_{\mathbf{p}, \text{RGB}}^k = \text{clip}_{[0,1]}(\exp(-\alpha_{\text{rgb}} e_{\text{RGB}}^k(\mathbf{p})) \cdot (1 - \lambda_{\text{sat}} \text{sat}^k(\mathbf{p}))). \quad (13)$$

Let $g^k(\mathbf{p}) = \|\nabla \tilde{D}^k(\mathbf{p})\|_2$. We use depth gradients as a lightweight reliability proxy: large gradients often coincide with discontinuities where monocular depth and finite-difference normals are less stable. Depth reliability is

$$c_{\mathbf{p}, D}^k = \exp(-\gamma_D g^k(\mathbf{p})). \quad (14)$$

Normal confidence is initialized as

$$c_{\mathbf{p}, N}^{k,(0)} = \exp(-\gamma_N g^k(\mathbf{p})). \quad (15)$$

After the first fusion pass yields $\hat{N}_c^k$, we compute the angular discrepancy

$$\theta^k(\mathbf{p}) = \arccos(\text{clip}(\langle \tilde{N}_c^k(\mathbf{p}), \hat{N}_c^k(\mathbf{p}) \rangle, -1, 1)), \quad (16)$$

and update confidence by

$$c_{\mathbf{p}, N}^{k,(1)} = \exp(-\beta \theta^k(\mathbf{p}) / \theta_{\text{th}}). \quad (17)$$

The update is used to reweight Eq. (36) and to trigger at most one additional fusion pass. To avoid unstable feedback loops, the update is detached:

$$c_{\mathbf{p}, N}^k \leftarrow \text{stopgrad}(c_{\mathbf{p}, N}^{k,(1)}). \quad (18)$$

If disabled, $c_{\mathbf{p}, N}^k = c_{\mathbf{p}, N}^{k,(0)}$.

Let $\{V_{\text{lex}}^j\}_{j=1}^J$ be lexicon embeddings. If s exists,

$$\pi_j = \frac{\exp(\cos(V_S, V_{\text{lex}}^j) / \tau)}{\sum_{\ell=1}^J \exp(\cos(V_S, V_{\text{lex}}^\ell) / \tau)}, \quad c_S = \max_j \pi_j. \quad (19)$$

If s is absent, $c_S = 0$ and the text modality is disabled in Section 3.3.2.

3.3.2. Confidence-weighted cross-modal attention fusion

Fusion integrates RGB evidence, depth-derived normals, and an optional text prior into a unified representation for normal prediction. We separate reliability modeling (confidence pooling and modality weighting) from representation interaction (cross-modal attention), which improves robustness under specularities and weak texture.

$$F_{\text{RGB}}^k = E_{\text{RGB}}(I^k), \quad F_N^k = E_N(\tilde{N}_c^k), \quad F_S^k = \text{broadcast}(W_S V_S), \quad (20)$$

where E_{RGB} and E_N are lightweight encoders and W_S projects the 512-D text embedding to C channels. We flatten each feature map on an $h \times w$ grid into $D = hw$ tokens, denoted by $\tilde{F}_m^k \in \mathbb{R}^{D \times C}$. If s is absent, the S branch is not instantiated and $\mathcal{M}(s) = \{\text{RGB}, N\}$.

Token confidences are computed by pooling pixel confidences gated by $\mathbb{I}_{\text{valid}}^k$:

$$\tilde{c}_{t,m}^k = \frac{\text{AvgPool}_{h \times w}(c_{\mathbf{p},m}^k \cdot \mathbb{I}_{\text{valid}}^k(\mathbf{p}))}{\text{AvgPool}_{h \times w}(\mathbb{I}_{\text{valid}}^k(\mathbf{p})) + \epsilon_{\text{div}}}, \quad t \in \{1, \dots, D\}, m \in \{\text{RGB}, N\}. \quad (21)$$

If s exists, $\tilde{c}_{t,S}^k \equiv c_S$.

A modality mask $\mu_m(s) \in \{0, 1\}$ is defined with $\mu_S(s) = \mathbb{I}(s \text{ exists})$ and $\mu_{\text{RGB}} = \mu_N = 1$. Fusion weights are

$$\omega_{t,m}^k = \frac{\mu_m(s) \exp(\tilde{c}_{t,m}^k / \tau_\omega)}{\sum_{m' \in \{\text{RGB}, N, S\}} \mu_{m'}(s) \exp(\tilde{c}_{t,m'}^k / \tau_\omega) + \epsilon_{\text{div}}}. \quad (22)$$

We add a shared 2D sine-cosine positional encoding $P \in \mathbb{R}^{D \times C}$ to preserve spatial correspondence:

$$\tilde{F}_m^k \leftarrow \tilde{F}_m^k + P, \quad m \in \mathcal{M}(s). \quad (23)$$

Let $\text{MHA}(Q, K, V)$ denote multi-head attention. If s exists,

$$\begin{aligned} \tilde{F}_{\text{RGB} \leftarrow (N, S)}^k &= \text{MHA}(\tilde{F}_{\text{RGB}}^k, [\tilde{F}_N^k; \tilde{F}_S^k], [\tilde{F}_N^k; \tilde{F}_S^k]), \\ \tilde{F}_{N \leftarrow (\text{RGB}, S)}^k &= \text{MHA}(\tilde{F}_N^k, [\tilde{F}_{\text{RGB}}^k; \tilde{F}_S^k], [\tilde{F}_{\text{RGB}}^k; \tilde{F}_S^k]), \\ \tilde{F}_{S \leftarrow (\text{RGB}, N)}^k &= \text{MHA}(\tilde{F}_S^k, [\tilde{F}_{\text{RGB}}^k; \tilde{F}_N^k], [\tilde{F}_{\text{RGB}}^k; \tilde{F}_N^k]). \end{aligned} \quad (24)$$

Otherwise,

$$\begin{aligned} \tilde{F}_{\text{RGB} \leftarrow N}^k &= \text{MHA}(\tilde{F}_{\text{RGB}}^k, \tilde{F}_N^k, \tilde{F}_N^k), \quad \tilde{F}_{N \leftarrow \text{RGB}}^k \\ &= \text{MHA}(\tilde{F}_N^k, \tilde{F}_{\text{RGB}}^k, \tilde{F}_{\text{RGB}}^k). \end{aligned} \quad (25)$$

We compute fused tokens as a confidence-weighted mixture of attention-updated modality tokens:

$$\tilde{F}_{\text{fused}}^k(t) = \sum_{m \in \mathcal{M}(s)} \omega_{t,m}^k \tilde{F}_{m \leftarrow (-m)}^k(t), \quad F_{\text{fused}}^k = \text{reshape}(\tilde{F}_{\text{fused}}^k). \quad (26)$$

A lightweight head predicts refined camera-frame normals:

$$\hat{N}_c^k(\mathbf{p}) = \text{normalize}(g_\phi(F_{\text{fused}}^k(\mathbf{p}))), \quad \hat{N}_c^k \in \mathbb{R}^{H \times W \times 3}. \quad (27)$$

Table 2
Stage-wise test-time refinement schedule.

Stage	Vars.	Losses
1	$\mathcal{V}$	$\lambda_{\text{photo}} \mathcal{L}_{\text{photo}} + \lambda_{\text{fp}} \mathcal{L}_{\text{fp}}$
2	$\mathcal{V}$	$\lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \lambda_{\text{fp}} \mathcal{L}_{\text{fp}}$
3	$\mathcal{V}$	$\lambda_{\text{normal}} \mathcal{L}_{\text{normal}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \lambda_{\text{fp}} \mathcal{L}_{\text{fp}}$
4	$\mathcal{T}$	$\lambda_{\text{color}} \mathcal{L}_{\text{color}}$

3.4. Mesh reconstruction and multi-stage refinement

3.4.1. Mesh initialization

We back-project aligned depth to world coordinates and attach refined normals:

$$\mathbf{x}_w^k(\mathbf{p}) = (\mathbf{R}^k)^\top (\mathbf{x}_c^k(\mathbf{p}) - \mathbf{t}^k), \quad (28)$$

$$\hat{\mathbf{n}}_w^k(\mathbf{p}) = (\mathbf{R}^k)^\top \hat{\mathbf{N}}_c^k(\mathbf{p}). \quad (29)$$

Samples are retained using valid depth and $c_{\mathbf{p},N}^k > \eta_N$. The oriented point cloud is aggregated across views and reconstructed by screened Poisson to obtain M_{init} [20].

3.4.2. Four-stage test-time mesh refinement

Refinement starts from M_{init} and optimizes vertex positions $\mathcal{V}$ and then appearance $\mathcal{T}$ under fixed topology $\mathcal{F}$. All losses are evaluated using differentiable rendering with a z-buffer visibility mask. For each view k , the renderer outputs a depth map $Z_M^k(\mathbf{p})$ and a camera-frame normal map $N_{M,c}^k(\mathbf{p})$ of the current mesh.

We use a stage-wise schedule to stabilize optimization under limited and noisy multi-view cues. Stage 1 anchors geometry to image evidence via flow-guided photometric alignment; Stage 2 enforces depth reprojection consistency; Stage 3 corrects normal directions while preserving sharp features under feature-preserving regularization; Stage 4 fits appearance after geometry is stabilized. Table 2 lists optimized variables and active losses at each stage. In all experiments we use fixed iteration budgets T_1, T_2, T_3 (Table A.10), with early stopping enabled in Stages 1 and 3 when relative loss change falls below ϵ_1 and ϵ_3 , respectively (Table A.10).

The total objective is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{photo}} \mathcal{L}_{\text{photo}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \lambda_{\text{normal}} \mathcal{L}_{\text{normal}} + \lambda_{\text{fp}} \mathcal{L}_{\text{fp}} + \lambda_{\text{color}} \mathcal{L}_{\text{color}}. \quad (30)$$

3.4.3. Stage-1: Optical flow refinement via photometric alignment

Dense flow $f_{i \rightarrow j}$ is computed between view pairs (i, j) . Let S_{vis}^i denote surface samples on faces visible in view i . For a sample $\mathbf{s}$,

$$\mathbf{p}_i = \Pi(\mathbf{K}(\mathbf{R}^i \mathbf{s} + \mathbf{t}^i)). \quad (31)$$

$$\mathbf{p}_j^* = \mathbf{p}_i + f_{i \rightarrow j}(\mathbf{p}_i). \quad (32)$$

Forward-backward consistency is enforced by

$$\mathbb{I}_{\text{fb}}(\mathbf{p}_i, \mathbf{p}_j^*) = \mathbb{I}(\|f_{i \rightarrow j}(\mathbf{p}_i) + f_{j \rightarrow i}(\mathbf{p}_j^*)\|_2 < \delta_{\text{fb}}). \quad (33)$$

The photometric loss is

$$\begin{aligned} \mathcal{L}_{\text{photo}} = & \sum_{(i,j) \in \mathcal{P}} \sum_{\mathbf{s} \in S_{\text{vis}}^i} \mathbb{I}_{\text{in}}(\mathbf{p}_i) \mathbb{I}_{\text{in}}(\mathbf{p}_j^*) \cdot \mathbb{I}_{\text{fb}}(\mathbf{p}_i, \mathbf{p}_j^*) \cdot c_{\mathbf{p}_i, \text{RGB}}^i \cdot \\ & c_{\mathbf{p}_j^*, \text{RGB}}^j \cdot \rho(\|I^i(\mathbf{p}_i) - I^j(\mathbf{p}_j^*)\|_1). \end{aligned} \quad (34)$$

3.4.4. Stage-2: Mesh geometric optimization via depth reprojection consistency

$$\mathcal{L}_{\text{depth}} = \sum_k \sum_{\mathbf{p}} \mathbb{I}_{\text{vis}}^k(\mathbf{p}) \cdot \mathbb{I}_{\text{valid}}^k(\mathbf{p}) \cdot c_{\mathbf{p},D}^k \cdot \rho(Z_M^k(\mathbf{p}) - \bar{D}^k(\mathbf{p})). \quad (35)$$

3.4.5. Stage-3: Normal direction correction

$$\mathcal{L}_{\text{normal}} = \sum_k \sum_{\mathbf{p}} \mathbb{I}_{\text{vis}}^k(\mathbf{p}) \cdot \mathbb{I}_{\text{valid}}^k(\mathbf{p}) \cdot c_{\mathbf{p},N}^k \cdot \left(1 - \langle N_{M,c}^k(\mathbf{p}), \hat{N}_c^k(\mathbf{p}) \rangle\right). \quad (36)$$

3.4.6. Feature-preserving regularization

Let e be an edge with dihedral angle ϑ_e . ϑ_e^0 is computed on M_{init} and

$$\mathbb{I}_{\text{sharp}}(e) = \mathbb{I}(|\pi - \vartheta_e^0| > \vartheta_{\text{sharp}}). \quad (37)$$

Edge weights are assigned by

$$w_e = w_{\text{sharp}} \cdot \mathbb{I}_{\text{sharp}}(e) + w_{\text{smooth}} \cdot (1 - \mathbb{I}_{\text{sharp}}(e)), \quad 0 < w_{\text{smooth}} \ll w_{\text{sharp}}, \quad (38)$$

and the penalty is

$$\mathcal{L}_{\text{fp}} = \sum_{e \in \mathcal{E}} w_e (\vartheta_e - \vartheta_e^0)^2. \quad (39)$$

3.4.7. Stage-4: Multi-view color projection

$$\mathbf{c}_{w,i} = -(\mathbf{R}^i)^\top \mathbf{t}^i, \quad (40)$$

$$\mathbf{d}_{w,i}(\mathbf{s}) = \text{normalize}(\mathbf{c}_{w,i} - \mathbf{s}), \quad (41)$$

We use a view-angle term $\kappa_i(\mathbf{s})$ to downweight grazing views; in practice we set $\kappa_i(\mathbf{s}) = \max(0, \langle \mathbf{n}_M(\mathbf{s}), \mathbf{d}_{w,i}(\mathbf{s}) \rangle)$, where $\mathbf{n}_M(\mathbf{s})$ is the current mesh normal at $\mathbf{s}$ in the world frame.

We denote by $\mathbb{I}_{\text{vis}}^i(\mathbf{s})$ whether surface sample $\mathbf{s}$ is visible in view i ; this is obtained by querying the renderer at $\mathbf{p}_i(\mathbf{s})$ and the corresponding z-buffer depth.

$$\tilde{w}_i(\mathbf{s}) = \mathbb{I}_{\text{vis}}^i(\mathbf{s}) \cdot \mathbb{I}_{\text{in}}(\mathbf{p}_i(\mathbf{s})) \cdot c_{\mathbf{p}_i(\mathbf{s}), \text{RGB}}^i \cdot \kappa_i(\mathbf{s}), \quad Z(\mathbf{s}) = \sum_j \tilde{w}_j(\mathbf{s}), \quad (42)$$

$$\mathbb{I}_{\text{col}}(\mathbf{s}) = \mathbb{I}(Z(\mathbf{s}) > \epsilon_{\text{div}}), \quad w_i(\mathbf{s}) = \frac{\tilde{w}_i(\mathbf{s})}{Z(\mathbf{s}) + \epsilon_{\text{div}}}. \quad (43)$$

The supervision color is the confidence-weighted aggregation from posed views:

$$\mathbf{C}^*(\mathbf{s}) = \sum_i w_i(\mathbf{s}) I^i(\mathbf{p}_i(\mathbf{s})). \quad (44)$$

$$\mathcal{L}_{\text{color}} = \sum_{\mathbf{s} \in \mathcal{S}} \mathbb{I}_{\text{col}}(\mathbf{s}) \cdot \|C_M(\mathbf{s}; \mathcal{T}) - \mathbf{C}^*(\mathbf{s})\|_1. \quad (45)$$

3.5. Training and inference procedure

Algorithm 1 summarizes the overall training and inference pipeline under the fixed camera protocol (Table A.10).

We fine-tune the diffusion backbone using Eq. (4). The fusion encoders and the normal head are trained with normal supervision and multi-view consistency, and the one-step confidence update in Eq. (17) is applied with stop-gradient as in Eq. (18) to avoid unstable feedback.

Given I^0 , we generate posed views $\{I^k\}_{k=1}^{N_v}$ and raw depths $\{D_{\text{raw}}^k\}_{k=1}^{N_v}$, solve per-view affine depth alignment to obtain $\{\bar{D}^k\}_{k=1}^{N_v}$ using Eq. (7), and compute $\{\hat{N}_c^k\}$ via Eq. (10). We then compute confidences and run fusion to obtain $\{F_{\text{fused}}^k\}$ and $\{\hat{N}_c^k\}$, reconstruct M_{init} by screened Poisson, and apply the staged refinement in Table 2 to obtain $\hat{M}$.

Reproducibility settings and hyperparameters are summarized in Appendix, including the fixed camera protocol and training/inference hyperparameters. Unless otherwise stated, we follow these settings for all experiments.

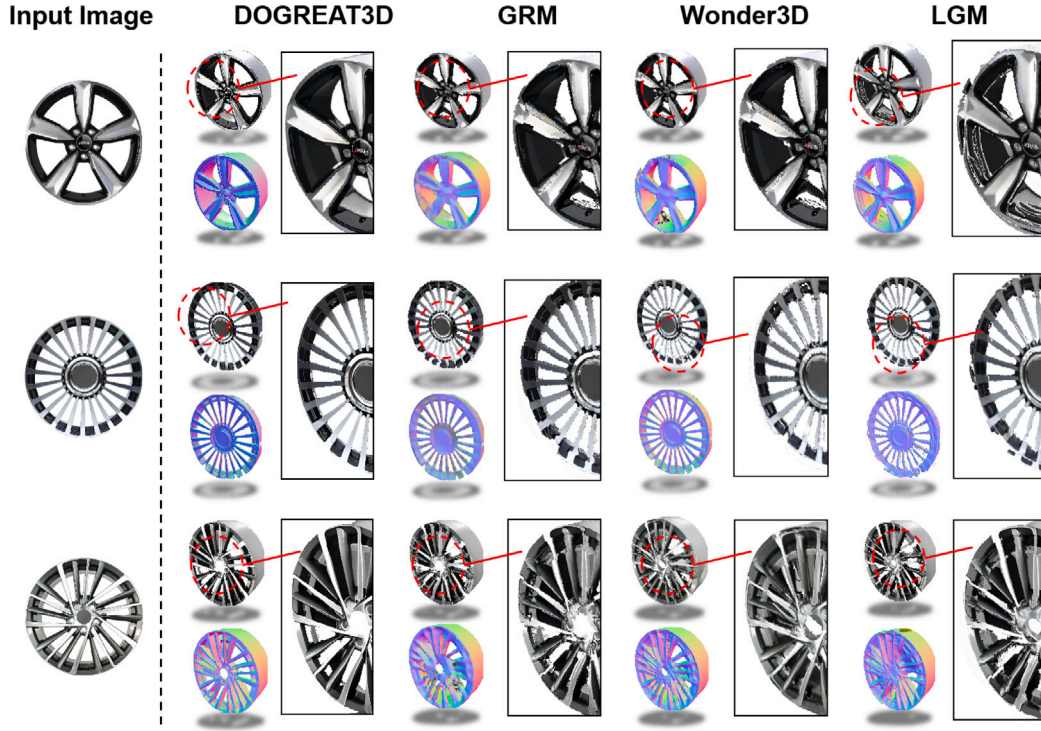


Fig. 3. Comparison of DOGREAT3D and baseline models on the wheel-hub 3D reconstruction task. Insets highlight thin spokes and hub openings.

Algorithm 1 DOGREAT3D: training and inference pipeline

Require: Training set with paired CAD geometry and rendered multi-view supervision; optional industrial prompts.

Require: Fixed camera protocol (Table A.10): intrinsics $\mathbf{K}$ and orbit poses $\{\mathbf{E}^k\}_{k=1}^{N_v}$.

Ensure: A trained diffusion backbone and fusion network; an inference procedure producing a textured mesh $\hat{M}$ from a single image.

- 1: **Training**
- 2: Fine-tune the diffusion backbone using Eq. (4) under the fixed camera protocol.
- 3: Train the confidence-aware fusion network with normal supervision and multi-view consistency.
- 4: **Inference** (given I^0 and optional prompt s)
- 5: Generate posed views and raw depths $\{I^k, D_{\text{raw}}^k\}_{k=1}^{N_v}$ conditioned on $\{\mathbf{E}^k\}$ and $\mathbf{K}$.
- 6: Estimate cross-view correspondences (optical flow) and align depths to obtain $\{\bar{D}^k\}_{k=1}^{N_v}$ by solving Eq. (7).
- 7: Compute depth-derived normals $\{\tilde{N}_c^k\}_{k=1}^{N_v}$ using Eq. (10).
- 8: Compute modality confidences and run fusion to obtain $\{\tilde{N}_c^k, F_{\text{fused}}^k\}_{k=1}^{N_v}$, with at most one confidence update.
- 9: Initialize a watertight mesh M_{init} by screened Poisson reconstruction.
- 10: Refine geometry and appearance under fixed topology using the staged schedule (Table 2) to obtain $\hat{M}$.

4. Experiments

4.1. Experimental setup

We evaluate DOGREAT3D on (i) WheelHub-1500, a CAD-paired industrial dataset curated for hub-like mechanical parts, and (ii) Google Scanned Objects (GSO), a general-domain dataset used to assess transfer under domain shift. WheelHub-1500 contains 1500 distinct parts,

Table 3

WheelHub-1500 dataset statistics (object-level, $n = 1500$).

Statistic	Mean	Median	P10	P90
#Faces (triangulated GT)	113 790	89 342	36 773	217 740
#Vertices (triangulated GT)	65 827	51 472	21 416	124 349
Sharp-edge ratio (%)	14.3	13.0	5.1	24.8
Silhouette hole count (input view)	11.9	10.5	4.0	23.0

Sharp-edge ratio fraction of edges whose dihedral angle θ_e on the triangulated GT satisfies $|\pi - \theta_e| > \theta_{\text{sharp}}$, using the same threshold as Section 3.4.6

Silhouette hole count number of holes in the binary silhouette of the conditional input view, used as a proxy for opening/spoke complexity.

each paired with a ground-truth CAD/mesh model in metric units. We use a fixed 70%/15%/15% train/validation/test split at the *object* level, ensuring that no part instance appears in multiple splits. Ground-truth geometry is used only for training supervision and quantitative evaluation; at inference, all methods take a single RGB image as input.

All WheelHub inputs are *rendered* from the paired CAD models under a controlled camera protocol with calibrated intrinsics and known orbit poses. Unless otherwise stated, we use $N_v = 8$ views with input resolution $H \times W = 512 \times 512$, orbit radius $r = 2.0$, elevation 10° , and azimuths $\{0, 45, \dots, 315\}^\circ$, consistent with Table A.10. To reflect realistic imaging variability and mitigate overfitting to a single render condition, we randomize illumination, background, and material parameters during rendering while keeping the camera protocol fixed. Table 3 summarizes the geometric complexity of WheelHub-1500 using triangulated GT statistics and two engineering-relevant proxies: sharp-edge ratio (Section 3.4.6) and silhouette hole count.

We use the standard GSO evaluation split solely for evaluation and do not train or fine-tune any component on GSO. Because metric scale is not guaranteed in GSO, all GSO geometry metrics are computed after similarity alignment (Section 4.1). We omit text prompts on GSO and evaluate in the geometry-first setting, mirroring typical single-image reconstruction usage where semantic prompting is unavailable or unreliable.

Although DOGREAT3D takes a single RGB image as input at inference, WheelHub-1500 provides ground-truth 3D geometry, enabling

Table 4

Quantitative comparison on the GSO evaluation split and WheelHub test split (all methods fine-tuned on WheelHub). Lower is better for CD/LPIPS; higher is better for IoU/F-score/PSNR/SSIM.

Method	GSO evaluation split						WheelHub test split					
	CD↓	IoU↑	F-Score↑	LPIPS↓	PSNR↑	SSIM↑	CD↓	IoU↑	F-Score↑	LPIPS↓	PSNR↑	SSIM↑
One-2-3-45	0.0313	0.4142	0.5518	0.1812	16.1058	0.8874	0.0943	0.2043	0.2636	0.2814	8.8238	0.4865
Wonder3D	0.0261	0.4663	0.6016	0.1536	18.0932	0.8995	0.0750	0.2331	0.2985	0.2746	9.7294	0.5321
Shap-E	0.0436	0.3584	0.4951	0.2023	15.6628	0.8621	0.0610	0.2150	0.2971	0.2832	10.1808	0.5604
LGM	0.0294	0.4792	0.6035	0.1678	17.8024	0.9032	0.0397	0.3115	0.3923	0.2181	12.4617	0.6322
GRM	0.0218	0.5167	0.6471	0.1523	18.9625	0.9187	0.0283	0.3617	0.4530	0.1904	14.2219	0.6890
Ours (DOGREAT3D)	0.0186	0.5264	0.6672	0.1428	24.2075	0.8953	0.0243	0.3682	0.5262	0.1758	20.3260	0.8044

Table 5

Fine-tuning and evaluation configurations for baselines on the WheelHub benchmark. “Orbit matched” indicates whether the method can be conditioned and evaluated under the same predefined orbit cameras.

Method	Input interface	Orbit matched	Fine-tuning data/signals on WheelHub	Notes
One-2-3-45	Single-view → multi-view	Yes	CAD-rendered RGB multi-view targets	View-conditioned synthesis
Wonder3D	Single-view → multi-view	Yes	CAD-rendered RGB multi-view targets	View-conditioned synthesis
Shap-E	Single-view → 3D	n/a	WheelHub training split under official recipe	Text branch disabled if unsupported
LGM	Single-view → 3D	n/a	WheelHub training split under official recipe	Same evaluation protocol
GRM	Single-view → 3D	n/a	WheelHub training split under official recipe	Strongest baseline in our setting

explicit multi-view supervision during training. For each training object, we render the ground-truth mesh under the same intrinsics and predefined orbit poses (Table A.10) to obtain multi-view RGB targets $\{I_{gt}^k\}_{k=1}^{N_v}$ and camera-frame normal maps $\{N_{gt,c}^k\}_{k=1}^{N_v}$. Each training sample is constructed by selecting one rendered view as the conditional input I^0 and treating the remaining views as pose-conditioned targets. This rendering-based protocol provides paired data for (i) diffusion backbone fine-tuning under Eq. (4) and (ii) supervised training of the fusion module and normal head under Eq. (27). At test time, the method receives only a single image (and an optional prompt), and all other views are synthesized without accessing any ground-truth geometry.

The multi-view generator is instantiated as Zero123Plus with ControlNet conditioning. We fine-tune the diffusion backbone on the WheelHub training split using the DDPM noise-prediction objective (Eq. (4)) with AdamW (learning rate 10^{-4} , weight decay 0.01, batch size 32), and apply early stopping based on validation loss (tolerance 10^{-4}). We then train the fusion encoders and normal head using rendered normal supervision and cross-view consistency under the same camera convention as Section 3. Test-time refinement does not backpropagate through the diffusion backbone and follows the staged schedule and hyperparameters in Tables 2 and A.10.

When the text modality is enabled, each WheelHub part is associated with a short industrial prompt describing the part type and salient attributes (e.g., spoke count and hub style). We use identical prompts for all methods that accept text conditioning; methods without a text interface are evaluated without text. To prevent prompt noise from harming geometry, DOGREAT3D gates the text branch by the global reliability score c_S (Eq. (19)) and treats text as an auxiliary modality in fusion (Section 3.3.2).

We compare against One-2-3-45 [33], Wonder3D [13], Shap-E [28], LGM [34], and GRM [35]. All baselines are executed using their official implementations when available. For WheelHub, we fine-tune each baseline on the same training split *through the training interface exposed by that method*. When a baseline supports view-conditioned synthesis, we match our orbit cameras and use CAD-rendered multi-view RGB supervision consistent with the baseline’s native training objective. When a baseline does not expose a compatible multi-view supervision interface, we follow its official fine-tuning recipe while holding the input resolution and evaluation protocol fixed. Table 5 summarizes the effective training interfaces and supervision signals used for each baseline.

We report both geometric and rendering metrics. **Geometry:** Chamfer Distance (CD) using 10,000-point surface sampling, Volume IoU (IoU), and F-score. **Rendering quality:** PSNR, SSIM, and LPIPS computed between rendered predictions and reference images under the

evaluation views. For WheelHub, ground-truth shapes are in metric units; we report CD in millimeters and use F-score@1 mm. For GSO, metric scale is not guaranteed; we perform similarity alignment for geometry evaluation and report F-score at a normalized threshold $\tau = 0.5\% \cdot \text{diag}(\text{bbox}_{gt})$, denoted as F-score@ τ . Unless otherwise stated, all methods are evaluated under the same orbit camera protocol and rendering settings.

For per-object comparisons, we conduct two-sided paired Wilcoxon signed-rank tests on CD, F-score, and LPIPS. Within each experimental block (baseline comparison and ablation tables), we apply Holm–Bonferroni correction and report corrected p -values when relevant. We summarize per-object variability using distribution plots when appropriate to distinguish consistent improvements from gains driven by a small subset of shapes.

We fix all random seeds to 42. For ablations and distribution analyses that require repeated runs, we sample a fixed subset of 30 WheelHub test parts once (seed = 42) and reuse it across all such experiments. Training is performed on 4× RTX 3090 GPUs (Ubuntu 20.04, CUDA 11.6, PyTorch 1.12).

Test-time inference is executed on a single NVIDIA RTX 4090. Table 6 reports the end-to-end inference time of DOGREAT3D with a stage-wise breakdown under the default setting (Table A.10); the average runtime is 75.1 s per object (range 48.2–106.5 s, $n = 30$), covering the full pipeline from view synthesis to final refined mesh.

4.2. Baseline comparison

Table 4 summarizes the aggregate performance on GSO and WheelHub. Qualitative comparisons of DOGREAT3D and the baseline methods are shown Fig. 3. We emphasize CD and F-score for geometric accuracy and completeness and LPIPS for perceptual similarity, and further report per-object distributions in Fig. 4 to assess the consistency of improvements.

On the WheelHub test split, DOGREAT3D achieves the best performance across all reported metrics in Table 4. Compared with the strongest baseline (GRM), DOGREAT3D reduces CD from 0.0283 mm to 0.0243 mm and increases F-score@1 mm from 0.4530 to 0.5262, while also improving LPIPS from 0.1904 to 0.1758. These gains are consistent with the design goal of mesh-level CAD-oriented reconstruction: confidence-aware fusion stabilizes surface orientation from depth-derived normals, and topology-preserving staged refinement sharpens thin spokes and openings without introducing topological artifacts.

To assess whether improvements are broad-based rather than driven by a few easy instances, we visualize per-object distributions (subset

Table 6

End-to-end inference time breakdown of DOGREAT3D on WheelHub (actual; $n = 30$ objects). Single NVIDIA RTX 4090, 512×512 , $N_v = 8$, default iteration budgets (T_1, T_2, T_3) and early stopping as in Table A.10.

Stage	Time (s) (mean $\pm$ std)	Range (min–max)
Pose-conditioned view synthesis (diffusion)	38.6 ± 6.5	27.3–52.8
Depth & optical flow estimation	7.2 ± 1.6	4.5–10.9
Depth alignment & normal cue derivation	6.8 ± 1.4	3.7–9.5
Confidence-aware fusion (normal prediction)	1.5 ± 0.4	0.7–2.8
Screened Poisson reconstruction (M_{init})	2.9 ± 0.7	1.4–4.8
Test-time refinement (Stages 1–3; geometry)	15.7 ± 3.8	9.2–23.6
Test-time refinement (Stage 4; color fitting)	2.4 ± 0.6	1.2–4.7
Total	75.1 ± 8.9	48.2–106.5

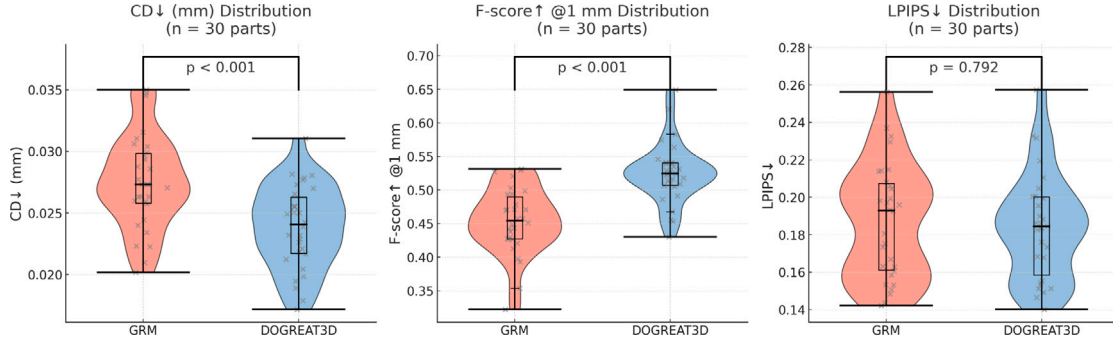


Fig. 4. Per-part distributions ($n = 30$) of $CD\downarrow$, $F\text{-score}\uparrow$, and $LPIPS\downarrow$ for DOGREAT3D versus GRM (both fine-tuned/evaluated on the WheelHub split as described in Table 5). Violin widths reflect KDE density; boxplots show median and interquartile range, and whiskers indicate the observed min–max. Cross markers denote individual parts. Corrected p -values (when reported) use paired Wilcoxon tests with Holm–Bonferroni correction within this block.

$n = 30$, seed = 42) for DOGREAT3D versus GRM in Fig. 4. Across CD, F-score, and LPIPS, DOGREAT3D exhibits a favorable shift in the distribution, with paired Wilcoxon tests (Holm–Bonferroni corrected within this block) indicating statistically significant improvements where reported.

On the GSO evaluation split, DOGREAT3D also improves geometry metrics relative to all baselines in Table 4, including CD (from 0.0218 to 0.0186 under similarity alignment) and F-score@ r (from 0.6471 to 0.6672). This indicates that the proposed geometric pipeline transfers beyond the industrial domain, despite all models being fine-tuned on WheelHub. For rendering metrics, DOGREAT3D achieves substantially higher PSNR and lower LPIPS, while SSIM is lower than GRM; in our setting SSIM is sensitive to local contrast and small spatial misregistrations introduced by view synthesis and rendering and does not necessarily track perceptual similarity. Accordingly, we treat LPIPS as the primary perceptual metric and report PSNR/SSIM as complementary indicators.

4.3. Ablation studies

All ablations are conducted on the fixed 30-part subset of the WheelHub test split (seed = 42). We report mean $\pm$ std across the 30 parts and perform paired tests against the corresponding reference configuration, with Holm–Bonferroni correction within each table. Qualitative ablation examples are provided in Fig. 5 to illustrate representative failure modes.

4.3.1. Modality contribution

We evaluate four configurations: **Full**, **–RGB**, **–Normals**, and **–Text**. All other components, including the refinement schedule and optimization hyperparameters, are kept unchanged. Quantitative results are reported in Table 7, with relative changes summarized in Fig. 6.

As shown in Table 7 and Fig. 6, removing RGB yields the largest degradation in geometry, increasing CD by 23.9% (relative) and decreasing F-score by 9.8% (relative; from 0.5262 to 0.4750, -0.0512

absolute). Removing normals also degrades geometry ($CD +21.4\%$ relative; $F\text{-score} -4.5\%$ relative, -0.0231 absolute), indicating that normal cues stabilize local surface orientation and fine detail. Removing text has a smaller effect on geometry but increases LPIPS (from 0.1758 to 0.1898; $+8.0\%$ relative), consistent with text acting primarily as an auxiliary semantic prior for appearance.

4.3.2. Fusion strategy

We compare three fusion strategies: **Concat** (channel concatenation), **Fixed-weight** (softmax-normalized static weights), and **Attention** (confidence-weighted cross-modal attention; Section 3.3.2). To isolate the fusion mechanism, we freeze the multi-view generator and train only the fusion module and normal head under the same data protocol. Results are reported in Table 8.

Attention yields the best overall trade-off. Compared with Attention, Concat increases CD and LPIPS while reducing F-score, indicating that naive fusion fails to adapt to modality reliability. Fixed-weight improves upon Concat but remains inferior to Attention, confirming that confidence-aware, sample-adaptive fusion is beneficial under specularities, low texture, and thin structures.

4.3.3. Test-time refinement stages

We ablate the refinement pipeline in Section 3.4.2 by progressively enabling its stages: **No refinement** (raw Poisson initialization M_{init}), **Only photometric** (Stage-1), **Add depth** (Stages 1+2), **Add normal** (Stages 1+2+3), and the **Full pipeline** (Stages 1–4). Quantitative results are reported in Table 9.

The refinement stages contribute cumulatively. Relative to the raw initialization, Stage-1 reduces CD and increases F-score, indicating that flow-guided photometric alignment improves cross-view consistency under the synthesized views. Adding Stage-2 further stabilizes geometry under depth reprojection constraints, and adding Stage-3 improves local surface orientation and detail recovery via normal alignment. Finally, Stage-4 improves PSNR/SSIM/LPIPS by fitting appearance only

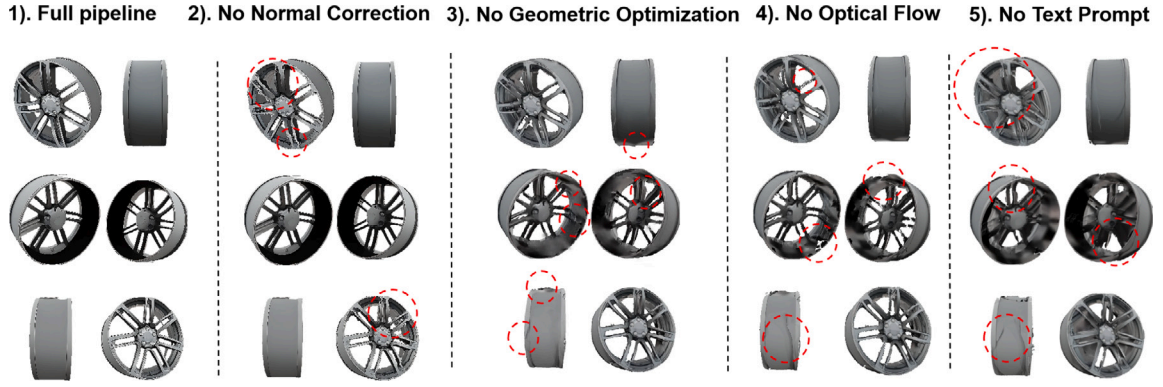


Fig. 5. Qualitative ablation results of DOGREAT3D with variants that remove or simplify key modules (e.g., normal correction and staged geometric refinement).

Table 7

Modality contribution ablation on the WheelHub subset ($n = 30$). Mean $\pm$ std are reported. Corrected p -values are from paired Wilcoxon tests with Holm–Bonferroni correction within this table.

Input configuration	CD↓	IoU↑	F-score↑	LPIPS↓	PSNR↑	SSIM↑	Corrected p -values (vs. Full)		
							CD	F-score	LPIPS
Full	0.0243 $\pm$ 0.0023	0.3682 $\pm$ 0.0301	0.5262 $\pm$ 0.0420	0.1758 $\pm$ 0.0142	20.3260 $\pm$ 0.9875	0.8044 $\pm$ 0.0223	–	–	–
–RGB	0.0301 $\pm$ 0.0035	0.3150 $\pm$ 0.0350	0.4750 $\pm$ 0.0500	0.1795 $\pm$ 0.0150	19.5421 $\pm$ 1.1200	0.7790 $\pm$ 0.0300	0.001	0.001	0.050
–Normals	0.0295 $\pm$ 0.0032	0.3375 $\pm$ 0.0280	0.5031 $\pm$ 0.0441	0.1782 $\pm$ 0.0148	20.0012 $\pm$ 0.9500	0.7915 $\pm$ 0.0205	0.003	0.005	0.200
–Text	0.0260 $\pm$ 0.0031	0.3621 $\pm$ 0.0305	0.5132 $\pm$ 0.0402	0.1898 $\pm$ 0.0167	20.1205 $\pm$ 1.0200	0.8032 $\pm$ 0.0198	0.020	0.010	0.045

Table 8

Fusion strategy ablation on the WheelHub subset ($n = 30$). Mean $\pm$ std are reported. Corrected p -values are from paired Wilcoxon tests with Holm–Bonferroni correction within this table.

Strategy	CD↓	IoU↑	F-score↑	LPIPS↓	PSNR↑	SSIM↑	Corrected p -values (vs. Attention)		
							CD	F-score	LPIPS
Concat	0.0280 $\pm$ 0.0028	0.3421 $\pm$ 0.0311	0.4901 $\pm$ 0.0452	0.1850 $\pm$ 0.0151	19.5421 $\pm$ 1.1200	0.7790 $\pm$ 0.0300	0.002	<0.001	0.020
Fixed-weight	0.0262 $\pm$ 0.0025	0.3578 $\pm$ 0.0295	0.5115 $\pm$ 0.0410	0.1800 $\pm$ 0.0143	20.1004 $\pm$ 1.0302	0.7950 $\pm$ 0.0140	0.012	0.005	0.080
Attention	0.0243 $\pm$ 0.0023	0.3682 $\pm$ 0.0301	0.5262 $\pm$ 0.0420	0.1758 $\pm$ 0.0142	20.3260 $\pm$ 0.9875	0.8044 $\pm$ 0.0223	–	–	–

Table 9

Refinement ablation on the WheelHub subset ($n = 30$). Mean $\pm$ std are reported. Corrected p -values (when shown) compare intermediate configurations against the Full pipeline, using paired Wilcoxon tests with Holm–Bonferroni correction within this table.

Configuration	CD↓	IoU↑	F-score↑	LPIPS↓	PSNR↑	SSIM↑	Corrected p -values (vs. Full)		
							CD	F-score	LPIPS
No refinement (M_{init})	0.0302 $\pm$ 0.0033	0.3203 $\pm$ 0.0352	0.4497 $\pm$ 0.0482	0.2001 $\pm$ 0.0182	18.9002 $\pm$ 1.1002	0.7602 $\pm$ 0.0253	–	–	–
Only photometric	0.0268 $\pm$ 0.0029	0.3453 $\pm$ 0.0322	0.4922 $\pm$ 0.0431	0.1897 $\pm$ 0.0161	19.3003 $\pm$ 1.0503	0.7798 $\pm$ 0.0232	0.005	0.008	0.020
Add depth	0.0257 $\pm$ 0.0026	0.3547 $\pm$ 0.0302	0.5123 $\pm$ 0.0412	0.1852 $\pm$ 0.0151	19.8002 $\pm$ 0.9903	0.7952 $\pm$ 0.0222	0.002	0.003	0.015
Add normal	0.0244 $\pm$ 0.0024	0.3653 $\pm$ 0.0292	0.5218 $\pm$ 0.0402	0.1799 $\pm$ 0.0141	20.1003 $\pm$ 0.9503	0.8002 $\pm$ 0.0212	0.0005	0.001	0.008
Full pipeline	0.0243 $\pm$ 0.0023	0.3682 $\pm$ 0.0301	0.5262 $\pm$ 0.0420	0.1758 $\pm$ 0.0142	20.3260 $\pm$ 0.9875	0.8044 $\pm$ 0.0223	–	–	–

after geometry converges, reducing the risk of geometry drift and reinforcing the stage-wise design in Table 2.

4.4. Robustness and sensitivity analyses

To test sensitivity to correspondence errors, we perturb the flow field used in depth alignment and Stage-1 by multiplicative scaling: $\tilde{f}(\mathbf{p}) = (1 + \delta)f(\mathbf{p})$ with $\delta \in \{-0.10, -0.05, 0, 0.05, 0.10\}$. We rerun refinement while keeping all other settings fixed. As shown in Fig. 7, performance degradation remains limited under moderate perturbations, indicating that confidence weighting and the multi-stage objectives mitigate correspondence noise.

We vary the iteration budget of geometry optimization (Stages 1–3) while keeping the stage ordering and all other hyperparameters fixed. Fig. 8 shows diminishing returns beyond a moderate budget, supporting the default configuration as a favorable efficiency–accuracy trade-off.

In practice, using 0.5 $\times$ to 1.0 $\times$ of the default Stage-1–3 iteration budget provides most of the achievable gains, while larger budgets yield marginal improvements.

4.5. Generalization to unseen industrial parts

We evaluate generalization on industrial parts outside the WheelHub distribution, including gear sets, guide rails, and thin-walled components. As shown in Fig. 9, DOGREAT3D preserves thin structures and sharp features despite domain shift, suggesting that feature-preserving refinement and confidence-aware fusion provide robust inductive biases for industrial geometry. These results also support the use of a fixed-topology refinement schedule in deployment scenarios where stability and repeatability are prioritized over unconstrained mesh editing.

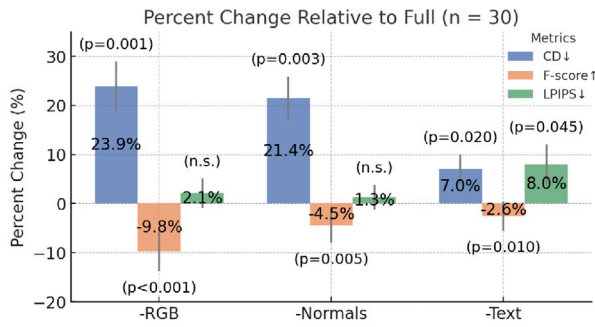


Fig. 6. Relative (%) change of CD, F-score, and LPIPS with respect to the full model on the WheelHub ablation subset. Bars indicate the mean relative change; error bars denote ± 1 standard deviation. Values shown are relative percentage changes, not percentage points.

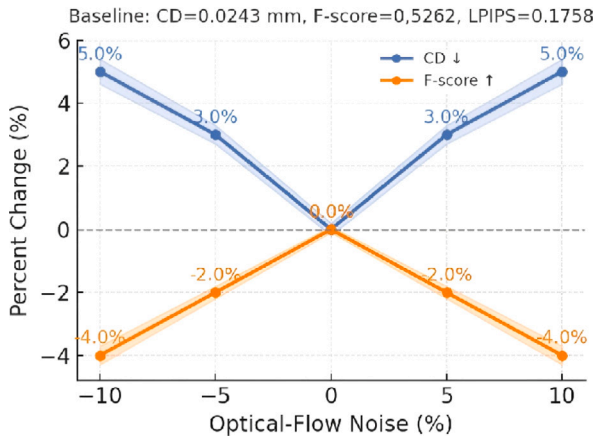


Fig. 7. Robustness to optical-flow perturbations on the WheelHub subset. We scale the flow field by $\tilde{f} = (1 + \delta)f$ with $\delta \in \{-10\%, -5\%, 0, +5\%, +10\%\}$ and report mean relative metric change (%) w.r.t. the unperturbed baseline; shaded regions denote ± 1 std.

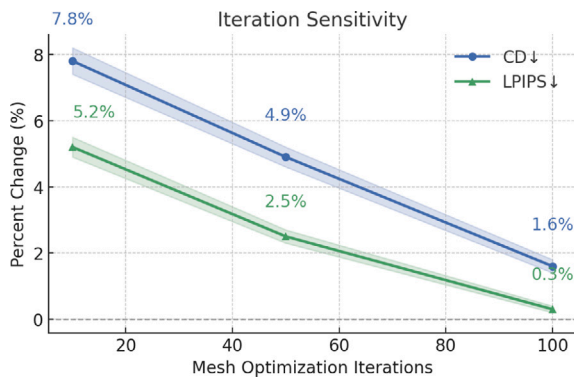


Fig. 8. Sensitivity to the optimization budget. We vary the iteration budget of the geometry stages (Stages 1–3) and report mean relative change (%) in CD and LPIPS w.r.t. the default setting; the curves show diminishing returns beyond moderate iteration counts.

4.6. Engineering case study: Linear static structural analysis

To assess downstream feasibility in a conventional CAE workflow, we perform a linear static analysis in ANSYS on a reconstructed WheelHub example. The goal is to demonstrate end-to-end workflow viability (geometry preparation $\rightarrow$ meshing $\rightarrow$ solver convergence) under

a simplified, reproducible load case, rather than certification-grade structural validation.

Starting from the reconstructed triangular mesh, we perform standard cleanup (removing isolated components, resolving non-manifold artifacts, and closing small spurious openings) to obtain a watertight surface for downstream preparation. We export the cleaned geometry to STEP for interoperability and define all quantities in a consistent mm–N–MPa unit system. To avoid scale ambiguity in CAE, we apply a single global scale prior to STEP export. For WheelHub, this scale is determined from the paired reference CAD model; in deployment, the same step can be replaced by a known physical measurement (e.g., outer diameter) or a calibrated imaging setup.

We assume a linear-elastic aluminum alloy with Young’s modulus $E = 69$ GPa and Poisson’s ratio $\nu = 0.33$. We constrain the hub at the mounting interface and apply a distributed radial load at the outer interface to emulate load transfer through the wheel assembly:

- **Constraint A** fixed support is applied to the hub mounting region (axle/bolt-circle interface)
- **Load A** total radial force of $F_r = 5$ kN is applied as a uniformly distributed load over the outer interface (rim-contact region), acting along the local radial direction toward the hub center

This load case is adopted as an engineering-motivated simplification and is not intended to replicate a specific standard test protocol.

The STEP solid is discretized using tetrahedral elements ($\sim 5 \times 10^5$ elements in our default setting), with local refinement around spoke–hub transitions and fillet regions. We solve a linear static problem in ANSYS using the standard direct sparse solver and default convergence criteria. Fig. 10 reports the total deformation field and confirms solver convergence under the above setup. Deformation concentrates around structurally sensitive transitions (e.g., spoke–hub junctions), illustrating that the reconstructed geometry can participate in a basic CAE analysis pipeline; the maximum total deformation is 0.00331 mm for the illustrated case.

5. Discussion

5.1. Summary of findings

The experiments show that DOGREAT3D improves geometric accuracy and completeness on the wheel-hub benchmark while maintaining competitive visual fidelity. Across both baseline comparisons and controlled ablations, the observed gains are consistent with the proposed design: (i) the cross-view depth alignment reduces geometric inconsistency under the predefined camera orbit, (ii) the confidence-aware fusion improves robustness when the synthesized views contain specular highlights or weak texture, and (iii) the CAD-oriented, topology-preserving test-time refinement sharpens features that are already supported by the initialization and multi-view evidence, without destabilizing the overall shape. Robustness results further suggest that moderate correspondence perturbations and reduced optimization budgets lead to bounded degradation, indicating a practical accuracy–runtime trade-off with the staged refinement schedule.

5.2. Engineering implications

Beyond standard reconstruction metrics, practical deployment benefits from outputs that can enter downstream CAD/CAE workflows as a high-quality mesh representation with limited manual intervention. The case study in Section 4.6 demonstrates workflow feasibility under our setup: after routine mesh cleanup and STEP export for interoperability, a reconstructed wheel-hub can be tetra-meshed and solved in ANSYS under a simplified load-transfer configuration, producing a stable deformation field. These results suggest that DOGREAT3D can provide an initial geometry for simulation-driven iterations when fast turnaround and approximate assessment are needed, while acknowledging that tolerance-critical decisions still require calibrated inputs and domain-specific validation.



Fig. 9. Reconstruction results of DOGREAT3D on unseen industrial parts (e.g., gear sets, guide rails, and thin-walled components).

Static structure
Type: Total deformation
Unit: mm
Time: 1

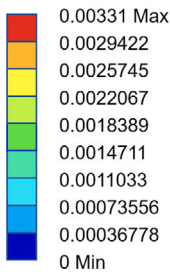


Fig. 10. ANSYS linear static analysis of a reconstructed wheel-hub model. Contour shows total deformation; the maximum displacement is 0.00331 mm.

5.3. Limitations

DOGREAT3D relies on a multi-view diffusion backbone to generate auxiliary views. Failure modes in view synthesis (e.g., hallucinated details, inconsistent textures, or viewpoint-dependent artifacts) may propagate to depth/flow estimation and consequently affect reconstruction fidelity. Although confidence modeling mitigates part of this issue, the pipeline remains sensitive to systematic synthesis bias.

Mesh refinement is performed under fixed topology, and the feature-preserving regularization primarily stabilizes and sharpens structures present in the Poisson initialization. As a result, missing topological elements or incorrect segmentation introduced early in the pipeline are difficult to recover during refinement. Moreover, while the refinement improves sharp-feature fidelity, it does not explicitly enforce parametric primitives, tolerance constraints, or manufacturability rules (e.g., minimum thickness and fillet standards).

For datasets without guaranteed metric scale (e.g., cross-domain evaluation), similarity alignment is necessary for geometry metrics, which limits conclusions about absolute dimensional accuracy. In engineering settings where absolute tolerances matter, additional scale calibration signals or reference measurements are required.

The structural analysis reported in Section 4.6 is a linear static case study intended to assess workflow compatibility. It does not include contact modeling, bolt preload, standardized load cases, or fatigue/impact evaluation. In addition, geometry repair and CAD conversion may alter small features that influence stress localization; therefore, the CAE results should be interpreted as feasibility evidence rather than certification.

5.4. Future work

Future work will focus on: (i) incorporating tolerance- and manufacturability-aware constraints during refinement (e.g., thickness regularization and primitive-aware fitting), (ii) improving multi-view synthesis and correspondence estimation under reflective materials via uncertainty modeling and view-consistency training, (iii) introducing lightweight scale calibration for real deployments using minimal reference measurements, and (iv) extending downstream validation to standardized structural load cases and fatigue/impact evaluation to better quantify engineering reliability.

6. Conclusion and future work

We introduced DOGREAT3D for CAD-oriented mesh reconstruction of industrial mechanical parts from a single RGB image. The pipeline combines viewpoint-conditioned multi-view synthesis under predefined camera poses, confidence-aware fusion of appearance and geometry cues, and a topology-preserving refinement schedule that sharpens engineering-critical features while maintaining shape stability.

On the wheel-hub benchmark, DOGREAT3D achieves higher geometric accuracy and completeness than the compared methods under the unified protocol in this paper. Against the strongest baseline, Chamfer Distance decreases by 14.1% (0.0283 mm to 0.0243 mm) and F-score@1 mm increases by 16.2% relative (0.4530 to 0.5262; 0.073 absolute), while perceptual quality remains comparable. On GSO, DOGREAT3D retains geometric gains (CD -14.7%) and improves F-score@ r by 3.1% relative (0.6471 to 0.6672; 0.020 absolute) under similarity alignment, indicating that the geometric components transfer beyond the industrial training distribution.

We also evaluated downstream usability. After routine mesh cleanup and STEP export for interoperability, a reconstructed wheel-hub can be tetra-meshed and solved in ANSYS under a simplified

Table A.10

Reproducibility settings and hyperparameters used in all experiments unless otherwise stated.

Item	Setting
Input resolution	$H \times W = 512 \times 512$ (all posed views); $I \in [0, 1]$ normalization; bilinear sampling at continuous p
#views N_v	8
Camera trajectory	orbit radius $r = 2.0$; elevation 10° ; azimuths $\{0, 45, \dots, 315\}^\circ$
Intrinsics $\mathbf{K}$	$c_x = W/2$, $c_y = H/2$; $f_x = f_y = \frac{0.5 \cdot W}{\tan(\text{FOV}/2)}$ with $\text{FOV} = 45^\circ$
View-pair set $\mathcal{P}$	Adjacent azimuth pairs (cyclic): $(k, k+1)$ and $(k, k-1)$
Depth estimator f_{depth}	Depth Anything V2-L (512 px inference), up-to-scale output D_{raw}^k
Flow estimator	RAFT [36] (512 px inference)
Depth alignment	3D-consistent affine per-view (a_k, b_k) with $a_k = \exp(\tilde{a}_k)$ via Eq. (7); gauge: $\tilde{a}_1 = 0, b_1 = 0$
Depth alignment regularization	$\lambda_a = 1.0 \times 10^{-3}$; $\lambda_b = 1.0 \times 10^{-3}$
Depth alignment optimizer	L-BFGS, 200 iters; init $\tilde{a}_k = 0, b_k = 0$; clamp $\epsilon_D = 10^{-3}$
Valid pixels $\Omega_{i \rightarrow j}$	in-bounds + finite aligned depth + valid flow + fwd-bwd $\delta_{\text{fb}} = 1.0$ px + photometric gate $\delta_I = 0.3$; drop saturated pixels via Eq. (8)
Charbonnier ρ (Eq. (11))	$\epsilon_\rho = 10^{-3}$; RGB uses $\rho(\ \Delta\mathbf{I}\ _1)$
RGB confidence (Eq. (13))	$\alpha_{\text{rgb}} = 5.0$; $\xi = 0.98$; $\lambda_{\text{sat}} = 0.5$; clip to $[0, 1]$
Depth reliability (Eq. (14))	$\gamma_D = 8.0$ (computed on aligned depth $\tilde{D}^k$)
Normal confidence init (Eq. (15))	$\gamma_N = 8.0$
Normal update (Eq. (17))	$\beta = 2.0$; $\theta_{\text{th}} = \pi/6$ (30°); stop-gradient enabled (Eq. (18))
Text relevance (Eq. (19))	$\tau = 0.07$; if no prompt: text modality masked ($\omega_{t,S} \equiv 0$); if prompt: $\tilde{c}_{t,S}^k \equiv c_S$
Token grid	$h = w = 64$ ($D = 4096$ tokens)
Feature channels	$C = 256$
Fusion temperature	$\tau_{\text{m}} = 0.2$ in Eq. (22)
Positional encoding	shared 2D sine-cosine $P \in \mathbb{R}^{D \times C}$ added to modality tokens (Eq. (23))
Attention	2 layers, 8 heads, head dim 32, dropout 0.0
Encoders E_{RGB}, E_N	lightweight ResNet-like (4 stages), output stride 8
Text projection W_S	linear $512 \rightarrow C$ then broadcast
Normal head g_ϕ	3-layer head, output 3 channels + ℓ_2 normalize (camera frame)
Poisson init	screened Poisson; depth = 10; samples/node = 1.5; trim = 0.0
Point filtering	$c_{p,N}^k > \eta_N$, $\eta_N = 0.3$; invalid depth removed
Renderer (differentiable)	PyTorch3D rasterizer; z-buffer visibility mask; bilinear sampling for images/confidences
Stage-1 sampling	$S'_{\text{vis}}: 20\text{k}$ triangle-area-proportional samples on visible faces per view; resample every 50 iters
Stage-1 correspondence gate	forward-backward consistency δ_{fb} via Eq. (33)
Stage-1 photometric weight	symmetric confidence: $c_{p,\text{RGB}}^i \cdot c_{p,\text{RGB}}^j$ (Eq. (34))
Numerical stability	$\epsilon_{\text{div}} = 10^{-8}$ for ratios; $\epsilon_D = 10^{-3}$ for depth positivity clamp
Stage-1 optimizer	Adam, lr = 1.0×10^{-3} , $T_1 = 300$ iters, early-stop $\epsilon_1 = 10^{-4}$
Stage-2 optimizer	Adam, lr = 5.0×10^{-4} , $T_2 = 500$ iters
Stage-3 optimizer	Adam, lr = 2.0×10^{-4} , $T_3 = 500$ iters, early-stop $\epsilon_3 = 10^{-4}$
Feature-preserving threshold	$\theta_{\text{sharp}} = 25^\circ$ (≈ 0.436 rad) (Eq. (37))
Feature-preserving weights	$w_{\text{sharp}} = 1.0$, $w_{\text{smooth}} = 0.1$ in Eq. (38)
Stage-4 sampling	$S: 50\text{k}$ triangle-area-proportional samples on the mesh; fixed barycentric coordinates during optimization
Stage-4 visibility/weighting	camera center $\mathbf{c}_{w,i} = -(\mathbf{R}^i)^T \mathbf{t}^i$ (Eq. (40)); supervise only if $Z(s) > \epsilon_{\text{div}}$ (Eq. (43))
Stage-4 (color)	geometry frozen; optimize per-vertex colors via Eq. (45) using $C^*(s)$ and view-angle κ_i
Loss weights	$\lambda_{\text{photo}} = 1.0$, $\lambda_{\text{depth}} = 1.0$, $\lambda_{\text{normal}} = 0.5$, $\lambda_{\text{fp}} = 0.1$, $\lambda_{\text{color}} = 0.2$
Hardware/runtime	1 $\times$ RTX 4090; 75.1 ± 8.9 s per object end-to-end (incl. refinement; see Table 6)

radial load-transfer setup, producing a stable deformation field. This case study is intended as workflow feasibility evidence rather than standardized structural certification, but it supports using DOGREAT3D outputs as an initial geometry for CAD/CAE iteration.

Performance depends on the cross-view consistency of synthesized views and the quality of correspondences; systematic artifacts can propagate through depth alignment, fusion, and refinement. Because refinement preserves topology, missing structures introduced during initialization are difficult to recover. When metric scale is not guaranteed, extra calibration is required for tolerance-sensitive applications. Finally, the CAE validation in this paper is limited to a linear static feasibility case.

Future work will prioritize improving cross-view consistency and uncertainty handling for reflective materials, adding manufacturability- and tolerance-aware constraints during refinement, introducing lightweight scale calibration for real deployments, and expanding downstream validation to standardized load cases and fatigue/impact evaluation.

CRediT authorship contribution statement

Xinxin Liang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Zuoxu Wang:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jihong Liu:** Writing – review & editing,

Writing – original draft, Supervision, Conceptualization. **Ang Liu:** Writing – review & editing, Supervision, Conceptualization. **Chun-Hsien Chen:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Reproducibility details

See Table A.10.

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